

Surgery vs. radiotherapy in patients with uveal melanoma

Analysis of the SEER database using propensity score matching and weighting

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Abstract

Background The treatment modalities for uveal melanoma (UM) include surgery and radiotherapy (RT). The utilization of RT as a strategy for organ preservation has been increasing, but the survival difference between the two aforementioned treatment modalities has not been reported.

Methods An observational and cohort study was performed using a propensity score with an already existing public database. Patients diagnosed with UM within the period from 2004–2013 were selected from the Surveillance, Epidemiology, and End Results (SEER) database. One-to-one matching and inverse probability of treatment weighting (IPTW) using the propensity score were used to estimate and compare survival rates.

Results Overall, 3291 patients were treated: 2503 received RT only (RT group) and 788 received surgical resection only (surgery group). The RT group had an improved crude 5-year overall survival (OS) rate compared with the surgery

group (76% vs. 60%, $P < 0.001$), and an improved 5-year melanoma-specific survival (MSS) rate (89% vs. 73%, $P < 0.001$). Compared to the surgery group, the RT group was associated with improved OS (hazard ratio [HR] 0.52, 95% confidence interval [CI] 0.38–0.73, $P < 0.001$) and MSS (HR 0.48, 95% CI 0.35–0.65, $P < 0.001$) in the matched cohort. The survival benefit of the RT group maintained after adjustment with IPTW, both in OS and MSS.

Conclusions To our knowledge, the present study was the first to demonstrate the survival difference between the two treatment modalities for UM using both the propensity score matching and weighting methods with the SEER database. The current study suggests that RT may provide a survival advantage over surgery in the treatment of UM.

Keywords Eye neoplasms · Overall survival · Brachytherapy · Eye enucleation · Organ preservation

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Chirurgie vs. Strahlentherapie bei Patienten mit Uveamelanom

Analyse der SEER-Datenbank mithilfe von Propensity-Score-Matching und -Gewichtung

Zusammenfassung

Hintergrund Die Behandlungsmodalitäten für das Uveame-lanom (UM) sind die Chirurgie und Strahlentherapie (RT). Die Nutzung der RT als Strategie zum Organerhalt hat zuge-nommen, aber der Unterschied in der Überlebensrate zwis-chen den beiden zuvor genannten Behandlungsmodalitäten wurde nicht berichtet.

Methoden Beobachtungs- und Kohortenstudie unter Ver-wendung eines Propensity-Scores mit einer bereits vor-handenen öffentlichen Datenbank. Patienten, bei denen im Zeitraum von 2004 bis 2013 ein UM diagnostiziert worden

war, wurden aus der Datenbank Surveillance, Epidemiology, and End Results (SEER) ausgewählt. Eins-zu-eins-Matching und inverse Behandlungswahrscheinlichkeitsgewichtung (IPTW) unter Heranziehung des Propensity-Scores wurden angewendet, um die Überlebensraten abzuschätzen und zu vergleichen.

Ergebnisse Insgesamt wurden 3291 Patienten behandelt: 2503 erhielten nur eine RT (RT-Gruppe) und 788 erhielten nur eine chirurgische Resektion (Chirurgiegruppe). Die RT-Gruppe hatte im Vergleich zur Chirurgiegruppe eine verbesserte 5-Jahres-Gesamtüberlebensrate (OS; 76 % vs. 60 %, $P < 0,001$) und eine verbesserte melanomspezifische 5-Jahres-Überlebensrate (MSS; 89 % vs. 73 %, $P < 0,001$). Im Vergleich zur Chirurgiegruppe war die RT-Gruppe mit einem verbesserten OS (Hazard Ratio [HR] 0,52, 95 %-Konfidenzintervall [KI] 0,38–0,73, $P < 0,001$ und MSS [HR 0,48, 95 %-KI 0,35–0,65, $P < 0,001$]) in der gematchten Kohorte assoziiert. Der Überlebensvorteil der RT-Gruppe blieb nach der Anpassung mit IPTW erhalten, sowohl im OS als auch im MSS.

Schlussfolgerungen Nach unserer Kenntnis war die vorliegende Studie die erste, die den Überlebensunterschied zwischen den beiden Behandlungsmodalitäten für UM unter Verwendung sowohl des Propensity-Score-Matchings als auch der Gewichtungsmethoden mit der SEER-Datenbank demonstrierte. Die aktuelle Studie deutet darauf hin, dass die RT einen Überlebensvorteil gegenüber der Operation in der Behandlung von UM bieten kann.

Schlüsselwörter Tumoren des Auges · Gesamtüberleben · Brachytherapie · Enucleatio bulbi · Organerhalt

Uveal melanoma (UM) is the most common primary ocular cancer for adults. It arises from the uveal tract, consisting of the choroid, ciliary body, and iris. The age-adjusted incidence of UM is 5.1 per million according to the Surveillance, Epidemiology, and End Results (SEER) program of the National Cancer Institute, remaining unchanged from 1973–2008 [1].

Surgery and radiotherapy (RT) are the major forms of treatment for UM. One of the surgical treatments is enucleation, which cannot conserve the affected eye and vision. Another surgical treatment is local tumor resection including eyewall resection or endoresection to conserve the globe. Adjuvant RT local resection is often performed to reduce recurrences.

Historically, RT has been an organ-preserving treatment, delivered by broadly two techniques: brachytherapy and external beam radiation therapy (EBRT). Brachytherapy using radioactive plaque has become a widely accepted treatment, with a few limitations, such as a large extraocular extension, optic nerve involvement, diffuse or multifocal tumor [2, 3]. The Collaborative Ocular Melanoma Study (COMS)

group conducted a randomized trial comparing brachytherapy with enucleation for the treatment of medium-sized choroidal melanoma in patients free of clinical evidence of metastasis and who were suitable candidates for either treatment [4]. The 12-year all-cause mortality was 43% in the brachytherapy arm and 41% in the enucleation arm. Based on the long-term results of the COMS trial, brachytherapy has been accepted to be an organ preservation method for UM with a good control rate, except in the case of a large tumor, a tumor with extraocular extension, or a painful eye [3]. However, there is criticism that the COMS trial lacks sufficient statistical power to prove that brachytherapy is better than enucleation in overall survival rates [5].

Recently, there has been a decrease in the number of surgeries performed and an increase in the number of RT-treated cases [1]. In addition, as the brachytherapy is a more labor-intensive and invasive procedure than EBRT [1], UM has been one of the indications of the EBRT techniques since the 1970s [6]. The clinical outcomes of EBRT, including charged-particle EBRT and stereotactic radiosurgery (SRS), demonstrated a good local-control rate (90–98.9%), similar to brachytherapy [7–10].

Although the COMS trial confirmed that there is no difference in survival between brachytherapy and enucleation for medium-sized choroidal melanoma, the differences in the clinical outcomes, including those of the other sizes or subsets of UM, remain unclear. Thus, re-evaluation is needed with the changes in the trend of the treatment modalities with the increased use of EBRT, including SRS.

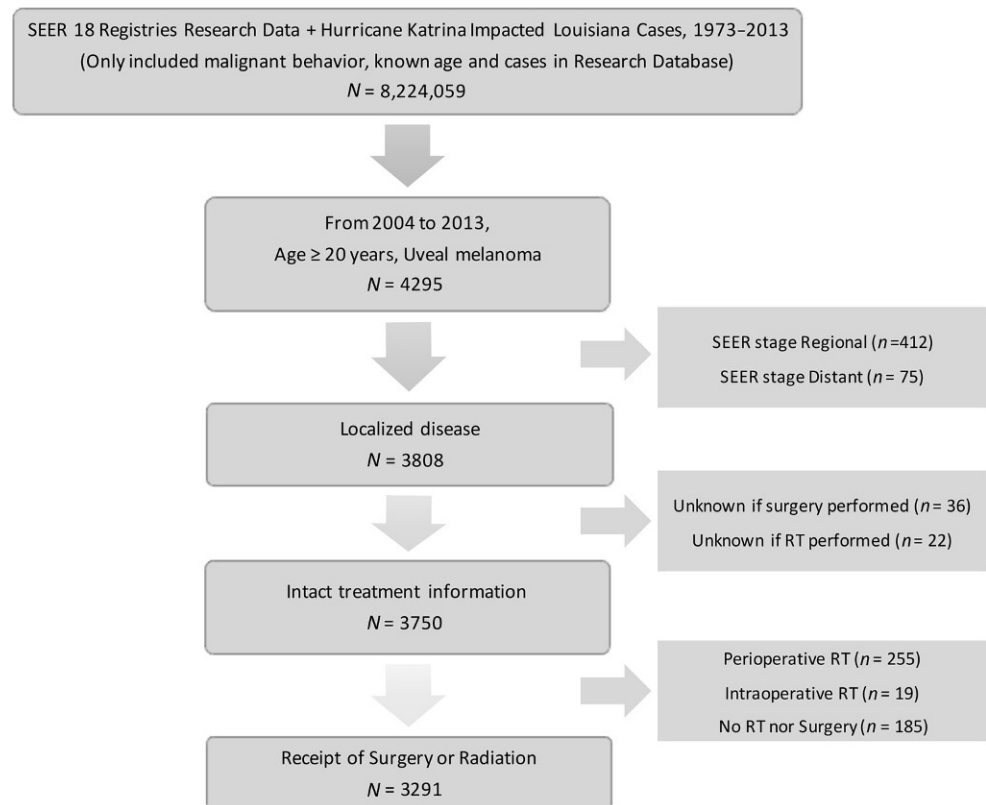
To investigate the clinical outcomes of groups receiving RT alone (RT group) and surgery alone (surgery group), respectively, patients diagnosed with UM during the period from 2004–2013 from the SEER database were retrospectively reviewed using propensity score matching and weighting.

Methods

Study population

The initial cohort was taken from the SEER 18 Registries (1973–2013), which covered approximately 27.8% of the U. S. population. Data were queried by using the SEER*Stat software (version 8.3.2). To obtain the study population, data were filtered and refined. First, the years of the diagnosis were limited to the period from 2004–2013 in an attempt to collect the site-specific factors coded since 2004. The patients whose age at diagnosis was equal to or greater than 20 years and whose diagnosis was uveal melanoma were all included, but the patients whose SEER stage was regional or distant upon diagnosis were excluded. Thus, to compare only two treatment modalities, all the cases of in-

Fig. 1 Study population diagram. Patient selection flowchart. *SEER* Surveillance, Epidemiology, and End Results, *RT* radiotherapy



traoperative RT, perioperative RT, no treatment, and absence of information on whether treatment was given were also excluded from the study population. A detailed description of patient selection is shown in Fig. 1.

Propensity score matching and inverse probability of treatment weighting

The propensity scores were estimated using the logistic regression model that had been established from the factors which potentially affected a decision of treatment modalities: RT versus surgery group. Those factors included gender, marital status, year of diagnosis, site of lesion, histology, and pathologic T stage. The size of the tumor was excluded because the matched cohort was not properly generated when it was considered; however, tumor size was when the inverse probability of treatment weighting (IPTW) cohort was constructed. With the estimated propensity score, a one-to-one matched cohort was constituted using the nearest-neighbor methods, within a 0.00039 caliper size [11].

The use of the IPTW method using propensity scores has increased of late [12]. It allows one to obtain weighted samples to examine without a confounding effect on the likelihood of specific treatments. The inverse probability treatment weights were calculated as follows: $1/(\text{propensity score})$ for the radiation group and $1/(1-\text{propensity score})$ for the surgery group [11]. The weights were normalized

to 1 [13]. The propensity scores were estimated from the logistic regression model using factors including gender, marital status, year of diagnosis, site of lesion, histology, pathologic T stage plus the size of the tumor. After the generation of the weights, they were used to weigh the study population so that survival analysis could be directly performed on it.

After matching or weighting, balance check was carried out on the characters between the surgery and RT groups. One of the existing balance check methods is calculating the standardized differences among the variables between groups [12]. As all the characters that were analyzed in the present study were dichotomous variables, the standardized differences were calculated using the prevalence of the variables in the radiation and surgery groups [14].

Statistical analysis

Overall survival (OS) and melanoma-specific survival (MSS) were estimated using the Kaplan–Meier method, and the log-rank test was carried out to compare the RT versus the surgery group. Chi-square (χ^2) tests were performed to identify the factors that contribute to the selection of each treatment modality. These factors were considered covariates and were incorporated into the models generating propensity scores. The Cox proportional hazard model adjusting all variables was constructed in the unmatched,

Table 1 Patient and tumor characteristics according to the treatment types

	Surgery group		RT group		Total		<i>P</i> -value
	<i>N</i>	%	<i>N</i>	%	<i>N</i>	%	
<i>Age (years)</i>							
<60	338	42.9	1071	42.8	1409	42.8	0.959
≥60	450	57.1	1432	57.2	1882	57.2	
<i>Gender</i>							
Women	340	43.1	1243	49.7	1583	48.1	0.001
Men	448	56.9	1260	50.3	1708	51.9	
<i>Marital status</i>							
Single	255	32.4	766	30.6	1021	31.0	<0.001
Married	436	55.3	1610	64.3	2046	62.2	
Unknown	97	12.3	127	5.1	224	6.8	
<i>Race</i>							
Caucasian	754	95.7	2410	96.3	3164	96.1	0.446
Non-Caucasian	34	4.3	93	3.7	127	3.9	
<i>Year of diagnosis</i>							
2004–2009	444	56.3	1301	52.0	1745	53.0	0.032
2010–2013	344	43.7	1202	48.0	1546	47.0	
<i>Site</i>							
Choroid	588	74.6	2293	91.6	2881	87.5	<0.001
Ciliary	182	23.1	184	7.4	366	11.1	
Iris	18	2.3	26	1.0	44	1.3	
<i>Histology</i>							
Malignant melanoma, NOS	317	40.2	2302	92.0	2619	79.6	<0.001
Mixed epitheloid/spindle	194	24.6	43	1.7	237	7.2	
Spindle	210	26.6	114	4.6	324	9.8	
Others ^a	67	8.5	44	1.8	111	3.4	
<i>Surgery type</i>							
Partial resection	120	15.2	0	0.0	120	3.6	<0.001
Enucleation/debulking	654	83.0	0	0.0	654	19.9	
NOS	14	1.8	0	0.0	14	0.4	
Not performed	0	0.0	2503	100.0	2503	76.1	
<i>Radiation type</i>							
Brachytherapy	0	0.0	1743	69.6	1743	53.0	<0.001
EBRT	0	0.0	719	28.7	719	21.8	
NOS	0	0.0	41	1.6	41	1.2	
Not performed	788	100.0	0	0.0	788	23.9	
<i>Size (diameter)</i>							
≤9 mm	145	18.4	624	24.9	769	23.4	0.001
>9 mm	343	43.5	1356	54.2	1699	51.6	
N/A	300	38.1	523	20.9	823	25.0	
<i>Thickness</i>							
≤6 mm	99	12.6	717	28.6	816	24.8	<0.001
>6 mm	120	15.2	289	11.6	409	12.4	
N/A	569	72.2	1497	59.8	2066	62.8	

Table 1 Patient and tumor characteristics according to the treatment types (Continued)

	Surgery group		RT group		Total		P-value
	N	%	N	%	N	%	
<i>T stage</i>							
TX	111	14.1	259	10.3	370	11.2	<0.001
T1	304	38.6	841	33.6	1145	34.8	
T2	201	25.5	1154	46.1	1355	41.2	
T3	111	14.1	190	7.6	301	9.1	
T4	61	7.7	59	2.4	120	3.6	
<i>Stage</i>							
I	286	36.3	800	32.0	1086	33.0	<0.001
II	254	32.2	1281	51.2	1535	46.6	
III	113	14.3	128	5.1	241	7.3	
Unknown	135	17.1	294	11.7	429	13.0	
Total	788	–	2503	–	3291	–	

P-value was estimated by χ^2 test

RT radiotherapy, EBRT external beam radiotherapy, NOS not otherwise specified, N/A not available

^aOther histology included nodular melanoma ($N = 1$), balloon cell melanoma ($N = 1$), amelanotic melanoma ($N = 23$), and epitheloid cell melanoma ($N = 86$)

matched, and IPTW-weighted cohorts. All the statistical analyses were performed using STATA version 14 (Stata-Corp, College Station, TX, USA) with the *psmatch2* and *pbalchk* packages.

Results

Patient and tumor characteristics & trends of treatment

Between 2004 and 2013, a total 3291 patients were selected from SEER registries in the present study. Of these, 788 (24%) patients were in the surgery group and 2503 (76%) patients were in the RT group. During the study period, patients treated with RT alone were increasing (6.9% for 2004 versus 13.1% for 2013), with corresponding wax and wane pattern in those treated with surgery alone (range 7.0–13.1%) (Supplementary Fig. 1a).

In both groups, the majority of treated site was choroid. The patients who had been diagnosed with T2 stage UM were more likely to undergo RT than surgery, while those who had been diagnosed with T1, T3, or T4 stage UM had a tendency to receive surgery more often than RT. Brachytherapy was carried out for more than two thirds of the patients (69.6%) in the RT group. This trend is depicted in Supplementary Fig. 1b and shows dominant use of brachytherapy over EBRT maintained over study period. Enucleation (83.0%) had a major role in the surgical treatment, and 15.2% of surgery group received partial resection. The dominant use of enucleation seems to be little difference in the tendency of the surgical method over study period (Supplementary Fig. 1c). The other histology of tumor (total $N = 111$) includes nodular melanoma ($N = 1$),

balloon cell melanoma ($N = 1$), amelanotic melanoma ($N = 23$), and epitheloid cell melanoma ($N = 86$). According to two groups, we compared the patient and tumor characteristics (Table 1). Gender, marital status, year of diagnosis, site of lesion, histology, size, and pathologic T stage was significantly different between the surgery and RT groups. Patients with tumor thickness ≤ 6 mm were more common in RT group than surgery group. In fact, 63% ($N = 2066$) of total patients had a missing value. Although tumor thickness seems to be differ between the surgery and RT groups, there is no significant distribution between groups ($P = 0.441$) as long as we only analyzed non-missing values.

Using the propensity score, the matched cohort and the weighted cohort using IPTW methods were constructed, respectively (Table 2). The matched cohort composed of 882 patients was obtained: 441 patients in the surgery and another 441 patients in the RT group. There was no significantly different distribution of clinical factors between the RT and the surgery group in the matched and the weighted cohorts. The detailed standardized differences are listed in Supplementary Table 1.

Overall survival

The 5-year OS rate was 76% for the RT group and 60% for the surgery group in the matched cohort ($P < 0.001$) (Fig. 1a) and 76% for the RT group and 50% for the surgery group in the weighted cohort ($P < 0.001$) (Fig. 1b), respectively. As can be seen in Table 3, the RT group maintained a lower risk of mortality compared to the surgery group in the fully adjusted model of the matched cohort (hazard ratio [HR] 0.52, 95% confidence interval [CI] 0.38–0.73, $P < 0.001$) and of the weighted cohort (HR 0.48, 95% CI

Table 2 Patient and tumor characteristics in the matched and the weighted cohort

	Matched cohort				<i>P</i> -value	IPTW cohort		
	Surgery group		RT group			<i>P</i> -value	Surgery group	RT group
	<i>N</i>	%	<i>N</i>	%			%	%
<i>Age (years)</i>								
<60	180	40.8	180	40.8	1.000	50.6	43.9	0.102
≥60	261	59.2	261	59.2		49.4	56.1	
<i>Gender</i>								
Women	193	43.8	200	45.4	0.635	51.7	47.9	0.345
Men	248	56.2	241	54.6		48.3	52.1	
<i>Marital status</i>								
Single	162	36.7	168	38.1	0.656	36.0	32.8	0.609
Married	242	54.9	243	55.1		58.5	61.5	
Unknown	37	8.4	30	6.8		5.5	5.6	
<i>Race</i>								
Caucasian	422	95.7	423	95.9	0.867	94.8	96.4	0.252
Non-Caucasian	19	4.3	18	4.1		5.2	3.6	
<i>Year of diagnosis</i>								
2004–2009	255	57.8	276	62.6	0.149	53.3	53.3	0.862
2010–2013	186	42.2	165	37.4		46.7	46.7	
<i>Site</i>								
Choroid	368	83.4	350	79.4	0.236	90.4	89.3	0.629
Ciliary body	68	15.4	87	19.7		9.7	10.7	
Iris	5	1.1	4	0.9		0.0	0.0	
<i>Histology</i>								
Malignant Melanoma, NOS	299	67.8	299	67.8	0.685	81.4	80.0	0.765
Mixed epitheloid/spindle	41	9.3	41	9.3		7.1	8.5	
Spindle	74	16.8	66	15		8.5	8.3	
Others ^a	27	6.1	35	7.9		3.0	3.2	
<i>Size (diameter)</i>								
≤9 mm	77	17.5	115	26.1	0.111	30.2	30.0	0.953
>9 mm	202	45.8	228	51.7		69.8	70.0	
N/A	162	36.7	98	22.2		0.0	0.0	
<i>T stage</i>								
TX	62	14.1	43	9.8	0.375	1.9	2.9	0.359
T1	155	35.1	155	35.1		27.4	31.2	
T2	146	33.1	158	35.8		55.7	49.2	
T3	57	12.9	63	14.3		11.1	12.2	
T4	21	4.8	22	5		3.9	4.5	
Total	441	–	441	–		100	100	

P-value was estimated by χ^2 test

RT radiotherapy, EBRT external beam radiotherapy, IPTW inverse probability of treatment weighting, NOS not otherwise specified, N/A not available

^aOther histology included nodular melanoma (*N* = 1), balloon cell melanoma (*N* = 1), amelanotic melanoma (*N* = 23), and epitheloid cell melanoma (*N* = 86)

0.35–0.65, *P* < 0.001). In addition, the histology of the spindle cell was associated with better OS in both the matched and the weighted cohort. Meanwhile, tumors larger than 9 mm was marginally associated with poorer OS compared to smaller tumors in the matched cohort (HR 1.50, 95% CI 1.00–2.25, *P* = 0.050). However, this tendency was not observed (adjusted hazard ratio [AHR] 1.67, 95% CI

0.98–2.85, *P* < 0.058) in the weighted cohort which had already been inversely weighted for the tumor size. The higher the T stage, the more detrimental effects on OS were observed in the both cohorts. When the survival outcomes were analyzed by T stage in the matched cohort (Supplement Fig. 2a), there was a significantly better OS in the T1 (*P* = 0.001) and the T2 stage (*P* < 0.001) in the RT group

Table 3 Cox proportional hazard regression models for overall survival

	Unadjusted (unmatched cohort, <i>N</i> = 3291)			Full-adjusted (unmatched cohort, <i>N</i> = 3291)			Full-adjusted (matched cohort, <i>N</i> = 882)			Full-adjusted (IPTW cohort)		
	HR	95% CI	<i>P</i> -value	HR	95% CI	<i>P</i> -value	HR	95% CI	<i>P</i> -value	AHR	95% CI	<i>P</i> -value
<i>Treatment</i>												
Surgery alone	Reference			Reference			Reference			Reference		
RT alone	0.56	0.48–0.65	<0.001	0.42	0.30–0.84	<0.001	0.52	0.38–0.73	<0.001	0.48	0.35–0.65	<0.001
<i>Age (years)</i>												
<60	Reference			Reference			Reference			Reference		
≥60	2.37	2.00–2.79	<0.001	1.39	1.08–1.79	0.011	2.00	1.41–2.84	<0.001	2.59	1.72–3.90	<0.001
<i>Gender</i>												
Female	Reference			Reference			Reference			Reference		
Male	1.17	1.01–1.36	0.034	1.14	0.88–1.47	0.322	1.13	0.81–1.58	0.463	0.90	0.63–1.27	0.541
<i>Marital status</i>												
Single	Reference			Reference			Reference			Reference		
Married	0.74	0.63–0.86	<0.001	0.90	0.69–1.19	0.460	0.81	0.58–1.13	0.216	0.76	0.52–1.12	0.161
Unknown	0.97	0.71–1.32	0.834	0.82	0.48–1.40	0.472	0.35	0.14–0.89	0.027	0.68	0.34–1.33	0.259
<i>Race</i>												
Caucasian	Reference			Reference			Reference			Reference		
Non-Caucasian	0.77	0.50–1.17	0.220	0.41	0.17–1.00	0.050	0.45	0.18–1.09	0.078	0.44	0.19–1.05	0.064
<i>Year of diagnosis</i>												
2004–2009	Reference			Reference			Reference			Reference		
2010–2013	1.18	0.96–1.44	0.114	0.90	0.58–1.39	0.637	0.79	0.41–1.51	0.482	1.07	0.55–2.08	0.851
<i>Site</i>												
Choroid	Reference			Reference			Reference			Reference		
Ciliary	1.33	1.09–1.62	0.006	1.41	0.98–2.03	0.068	1.33	0.87–2.03	0.185	1.15	0.78–1.69	0.481
Iris	0.17	0.02–1.20	0.076	NR			NR			NR		
<i>Histology</i>												
Malignant melanoma, NOS	Reference			Reference			Reference			Reference		
Mixed epithelioid/spindle	1.97	1.57–2.46	<0.001	1.45	0.97–2.18	0.072	1.31	0.84–2.04	0.234	1.27	0.89–1.81	0.184
Spindle	0.67	0.51–0.89	0.005	0.59	0.36–0.96	0.034	0.39	0.23–0.68	0.001	0.52	0.35–0.76	0.001
Others ^a	2.12	1.54–2.93	<0.001	0.69	0.33–1.45	0.330	1.36	0.79–2.35	0.270	1.00	0.61–1.66	0.988
<i>Size (mm)</i>												
≤9 mm	Reference			–			–			–		
>9 mm	1.87	1.50–2.32	<0.001	2.46	1.67–3.64	<0.001	1.50	1.00–2.25	0.050	1.67	0.98–2.85	0.058
<i>T stage</i>												
TX	1.52	1.17–1.99	0.002	2.86	1.21–6.79	0.017	0.73	0.10–5.41	0.759	1.48	0.80–2.75	0.212
T1	Reference			Reference			Reference			Reference		
T2	1.35	1.13–1.60	0.001	1.78	1.24–2.55	0.002	1.56	1.03–2.38	0.037	1.77	1.12–2.80	0.015
T3	2.75	2.05–3.67	<0.001	3.03	1.77–5.16	<0.001	2.35	1.24–4.45	0.009	2.13	1.27–3.58	0.004
T4	3.77	2.49–5.71	<0.001	5.02	2.44–10.31	<0.001	7.52	3.19–17.74	<0.001	2.60	0.97–7.02	0.059

P-value was estimated by χ^2 test

RT radiotherapy, IPTW inverse-probability treatment weighted, HR hazard ratio, AHR adjusted hazard ratio, CI confidence interval, NR not reported

^aOther histology included nodular melanoma (*N* = 1), balloon cell melanoma (*N* = 1), amelanotic melanoma (*N* = 23), and epithelioid cell melanoma (*N* = 86)

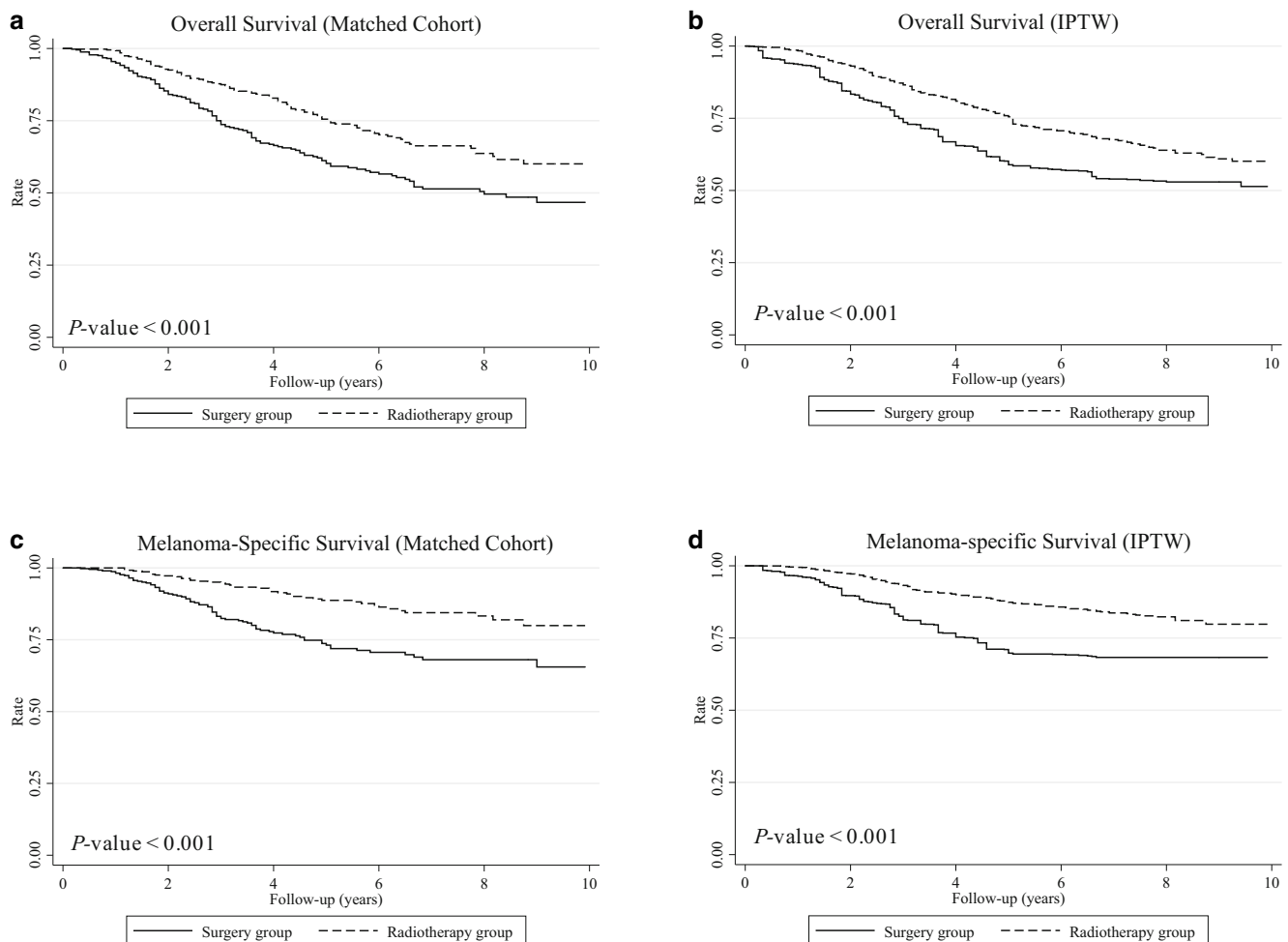


Fig. 2 Kaplan–Meier Curves and log-rank test for survival analysis. Comparing overall survival between the RT and surgery groups in the matched cohort (**a**) and the weighted cohort (**c**). Melanoma-specific survival was examined in the matched cohort (**b**) and the weighted cohort (**d**). P-value was estimated by log-rank test. RT radiotherapy

than in the surgery group. However, there were no significant differences in OS between the RT and the surgery groups in the T3 ($P = 0.360$) and T4 stages ($P = 0.220$). This trend remained in the weighted cohort as well (Supplement Fig. 2b).

Melanoma-specific survival

The 5-year MSS rate was 89% for the RT group and 73% for the surgery group ($P < 0.001$) in the matched cohort (Fig. 2c) and 87% for the RT group and 70% for the surgery group (Fig. 2d), respectively. In the fully adjusted model, RT is associated with a decreased risk of mortality compared to surgery (HR 0.35, 95% CI 0.22–0.56, $P < 0.001$) in the matched cohort; this trend was similar in the weighted cohort (AHR 0.35, 95% CI 0.23–0.52, $P < 0.001$) (Table 4). In addition, UM of spindle cell was significantly associated with improved MSS compared with other histologic types of UM. Patients with a tumor size of more than 9 mm had

significantly decreased MSS compared to those otherwise in the matched cohort (HR 2.81, 95% CI 1.41–5.59, $P = 0.003$), but not in the weighted cohort. In both the matched and the weighted cohort, RT group demonstrated a significantly better MSS in the T1 stage and the T2 stage as compared to the surgery group (Supplementary Fig. 3).

Discussion

To our knowledge, this study was the first to demonstrate the survival advantage of RT over surgery for the treatment of UM, especially in the early T stage. The clinical outcomes of 3291 patients treated with surgery or radiation for UM identified and obtained from the SEER database were examined. To overcome selection bias to decide treatment modalities (RT alone vs. surgery alone) from retrospective data, two major methods using the propensity score were

Table 4 Cox proportional hazard regression models for melanoma-specific survival

	Unadjusted (unmatched cohort, <i>N</i> = 3291)			Full-adjusted (unmatched cohort, <i>N</i> = 3291)			Full-adjusted (matched cohort, <i>N</i> = 882)			Full-adjusted (IPTW cohort)		
	HR	95% CI	<i>P</i> -value	HR	95% CI	<i>P</i> -value	HR	95% CI	<i>P</i> -value	AHR	95% CI	<i>P</i> -value
<i>Treatment</i>												
Surgery alone	Reference			Reference			Reference			Reference		
RT alone	0.46	0.37–0.56	<0.001	0.42	0.30–0.59	<0.001	0.35	0.22–0.56	<0.001	0.35	0.23–0.52	<0.001
<i>Age (years)</i>												
<60	Reference			Reference			Reference			Reference		
≥60	1.40	1.13–1.74	0.002	1.39	1.08–1.79	0.011	1.38	0.87–2.18	0.170	1.70	0.97–2.97	0.061
<i>Gender</i>												
Women	Reference			Reference			Reference			Reference		
Men	1.11	0.90–1.37	0.333	1.14	0.88–1.47	0.322	1.03	0.66–1.62	0.894	1.02	0.63–1.67	0.922
<i>Marital status</i>												
Single	Reference			Reference			Reference			Reference		
Married	0.85	0.68–1.06	0.151	0.90	0.69–1.19	0.460	0.99	0.62–1.57	0.950	0.66	0.38–1.16	0.147
Unknown	1.18	0.77–1.80	0.448	0.82	0.48–1.40	0.472	0.40	0.12–1.36	0.142	0.66	0.25–1.70	0.385
<i>Race</i>												
Caucasian	Reference			Reference			Reference			Reference		
Non-Caucasian	0.55	0.28–1.12	0.100	0.41	0.17–1.00	0.050	0.16	0.02–1.16	0.070	0.17	0.05–0.54	0.003
<i>Year of diagnosis</i>												
2004–2009	Reference			Reference			Reference			Reference		
2010–2013	1.14	0.85–1.51	0.382	0.90	0.58–1.39	0.637	0.64	0.26–1.61	0.345	0.95	0.38–2.35	0.910
<i>Site</i>												
Choroid	Reference			Reference			Reference			Reference		
Ciliary body	1.41	1.07–1.87	0.014	1.41	0.98–2.03	0.068	1.28	0.71–2.32	0.412	1.38	0.79–2.42	0.264
Iris	NR			NR			NR			NR		
<i>Histology</i>												
Malignant melanoma, NOS	Reference			Reference			Reference			Reference		
Mixed epitheloid/ spindle	2.35	1.74–3.16	<0.001	1.45	0.97–2.18	0.072	1.50	0.86–2.64	0.155	1.26	0.77–2.08	0.362
Spindle	0.67	0.45–1.00	0.051	0.59	0.36–0.96	0.034	0.30	0.13–0.67	0.003	0.48	0.28–0.82	0.008
Others ^a	1.95	1.21–3.14	0.006	0.69	0.33–1.45	0.330	0.77	0.30–1.97	0.585	0.57	0.23–1.38	0.212
<i>Size (mm)</i>												
≤9 mm	Reference			Reference			Reference			Reference		
>9 mm	3.21	2.23–4.62	<0.001	2.46	1.67–3.64	0.000	2.81	1.41–5.59	0.003	2.04	0.76–5.48	0.159
<i>T stage</i>												
TX	2.03	1.39–2.97	<0.001	2.86	1.21–6.79	0.017	1.31	0.17–10.02	0.796	2.74	1.30–5.78	0.008
T1	Reference			Reference			Reference			Reference		
T2	1.69	1.30–2.20	<0.001	1.78	1.24–2.55	0.002	1.55	0.86–2.82	0.147	2.21	1.05–4.68	0.037
T3	3.94	2.65–5.86	<0.001	3.03	1.77–5.16	<0.001	2.10	0.84–5.26	0.112	2.19	1.02–4.69	0.045
T4	5.67	3.29–9.77	<0.001	5.02	2.44–10.31	<0.001	7.99	2.39–26.72	0.001	3.75	1.11–12.68	0.033

P-value was estimated by χ^2 test

RT radiotherapy, IPTW inverse-probability treatment weighted, HR hazard ratio, AHR adjusted hazard ratio, CI confidence interval, NR not reported

^aOther histology included nodular melanoma (*N* = 1), balloon cell melanoma (*N* = 1), amelanotic melanoma (*N* = 23), and epitheloid cell melanoma (*N* = 86)

employed: the matching and the IPTW, which are commonly used in large observational studies.

In addition to the treatment modality, several factors are related to the survival outcome. The larger tumor diameter [15, 16], involvement of ciliary body [17], age [16], and extrascleral extension [15], and monosomy of chromosome 3 [18] are reported to be associated with a poor prognosis. In our study, the IPTW cohort with no different distribution of the tumor size between the RT and the surgery group was constructed, and the fully adjusted Cox hazard proportional model revealed that tumor size was marginally associated with overall survival. We only used 9 mm as the cut-off value of tumor size, which can divide small- and medium-sized tumors. Although we sought to evaluate the beneficial effect of RT over surgery even in large-sized tumors (>15–16 mm), approximately 38% of all patients were recorded at 10 mm intervals (e. g., 1–9 mm, 10–19 mm, 20–29 mm) and another 37% of all patients were at 3 mm intervals. Thus, we chose 9 mm of size as a cut-off in order to merge two populations, minimizing missing data about the tumor size. On the other hand, T stages reflecting extrascleral extension and ciliary body involvement as well as tumor size were used as a covariate to construct the matched and the IPTW cohorts in the present study. The prognostic importance of T stages is verified in terms of the 2010 AJCC staging system, demonstrating the higher risk of developing distant metastases as T stage increases [19]. Age and T stages were found to be significant risk factors for OS in the matched and the IPTW cohorts in our study, which is consistent with the result of the another SEER study that compared EBRT versus brachytherapy as treatment modalities for choroidal melanoma [20]. According to the modified Callender classification system, an increased portion of the epitheloid cell relative to the spindle cell in the mixed-cell type is associated with a worse prognosis [21, 22]. As expected, the histology of the spindle cell was linked to better OS and MSS rates. However, the worse prognosis of the mixed-cell type was not clearly identified in our study, probably because that the variability of the proportion of the epitheloid cells in the mixed-cell type had diluted the worse prognosis.

With regard to the site of the lesion, it does not seem to be associated with the OS or MSS rate. No difference in the OS rate stratified by the site of the lesion was found, except in the fully adjusted Cox proportional hazard model of the IPTW cohort. Because the incidence of iris melanoma was low, accounting for only 1.3% of the whole study population, the hazard ratio of iris melanoma cannot be estimated independently in most fully adjusted Cox proportional hazard models. Nevertheless, we can find no significant difference in survival rate between melanoma arising from the choroid and melanoma arising from the ciliary body.

As UM is limited within the globe where lymphatic vessels are not present, metastases are likely to develop hematogenous metastases [23]. The major metastatic site of UM is the liver, leading to a high incidence of mortality. Gragoudas et al. [24] reported that once metastasis develops, even though the patient received treatment for metastases, the 1-year OS rate is 19%. In this study, the patients treated with surgery showed worse clinical outcomes than those treated with RT, probably owing to distant metastasis, as can be seen in the MSS rate. Zimmerman et al. [25] assumed that the surgical procedure itself could lead to an increased risk of metastasis via the dissemination of the tumor emboli, which implies that the invasiveness of surgery may unintentionally increase the metastatic chances of the tumor. The detailed mechanisms, however, should be investigated in further studies. Furthermore, the present study only include a study population registered since 2004, when the COMS trial results were published.

Types of RT include EBRT and brachytherapy using radioactive plaque. EBRT can be delivered with charged particle such as protons or carbon ions as well as photons. Using charged particle irradiation, a local control rate of 95% at 15 years after proton beam therapy was reported [26]. A meta-analysis from 27 studies with 8809 UM patients concluded that the use of charged particle therapy was associated with the lower rate of local recurrence compared with brachytherapy [27]. Although comparing clinical outcomes among RT techniques to treat UM is beyond the scope of our study, charged particle therapy is likely to contribute to the improved OS or MSS rate over surgery alone.

Stratified by T stage, the RT group with early T stage UM showed significantly better OS and MSS rates in both cohorts. Meanwhile, there were no significant differences in the OS and MSS rates between the RT and surgery groups with T3 or T4 stages of UM. Thus, the clinical outcomes can be predicted, and the treatment modality can be determined depending on the T stage. Specifically, early T stage of UM can be managed with RT alone over surgery.

There might be the possibility of taking any secondary enucleation that may have been performed consequential to the brachytherapy. Fortunately, SEER data have recorded the sequence of treatments between RT and surgery. Therefore, we could exclude patients treated with combinational approaches according to SEER records.

This study has several limitations. First, the SEER database itself does not tell us what proportion of all UM treated in medical centers participating in SEER were registered. In the past, many patients were surgically treated by an ophthalmologist first without consultation to an oncologist. Thus, we selected only study patients from 2004–2013 to rule out missing patients that could affect our results. However, we also found that the use of radiotherapy was

increasing during that period. A previous study [1] demonstrating incidence, treatment, and survival with primary uveal melanoma in the United States over a 36-year period from 1973–2008 revealed that enucleation was the major treatment modality for treating UM before the COMS trial. Therefore, our results should be interpreted cautiously considering the changing trend of treatment modalities. Second, comparing types of RT (EBRT versus brachytherapy) and surgery (partial resection versus enucleation) is needed in another study as a separate analysis. Third, the SEER database does not record any information about the genetics of the UM such as monosomy of chromosome 3 or gene expression profiling which can define high-risk groups for development of metastatic disease. In addition, information about recurrence or toxicity is not available in the SEER database. Likewise, our statistical approach can only rule out factors available in the SEER database, harboring the risk of overlooking undocumented confounders. Therefore, a well-designed randomized controlled study is needed to verify and solidify our results favoring RT in the treatment of UM.

The present study, however, does provide strong support for the use of RT in early T stage UM due to its better clinical outcome over surgery. In addition, as the results of this study were based on the validated T staging of AJCC, they are reliable for use in predicting the survival rate clinically and are subsequently appropriate for use in patient consultation. Our results considered uncommon UM variants such as ciliary and iris melanoma, and patients treated with EBRT or local resection were also included in the study. Therefore, this study virtually provided a framework for RT as the first treatment modality choice for UM.

To our knowledge, there is currently no research showing clear benefit of RT over surgery for uveal melanoma. Our analysis based on the SEER database with two major methods using propensity scores demonstrated coherent survival benefits of radiation therapy in both the matched and the weighted cohorts. Therefore, RT including EBRT can be suggested as the first and definitive treatment of choice for uveal melanoma.

Compliance with ethical guidelines

Conflict of interest B.-S. Jang, J.H. Chang, S. Oh, Y.J. Lim and I.H. Kim declare that they have no competing interests.

Ethical standards All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. For this type of study formal consent is not required.

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